# Free fall

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Describe and calculate ideal free fall near Earth.

**SUCCESS CRITERIA - I can...**

1. State that you are ignoring air resistance.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “free fall” and “initial” to explain your answer. Sentence starters are optional.

**free fall**: motion with gravity as the only force

**initial**: at the beginning of the interval; subscript i

**acceleration**: rate of change of velocity

## 01 Warm-up: bring back an earlier idea

A **4 kg** cart starts from rest under **24 N** net force right for **3 s**.
1. **Find acceleration** and final velocity.
2. **State one thing you** are treating as true.

**Word help:** net force = the vector sum of all forces on one object

**Sentence starter:** I think … . I can point to …, which shows …, so … .

**Marzano check - circle:** ① ② ③

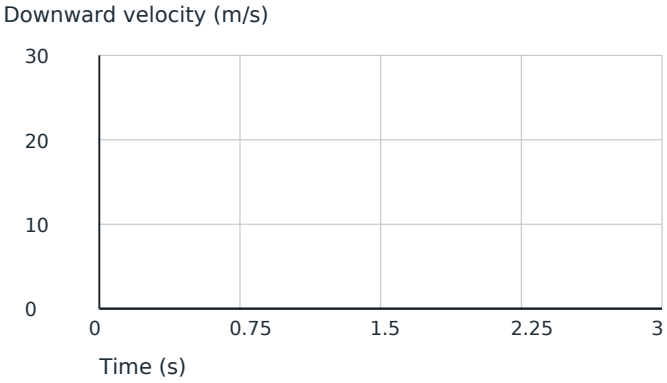
## 02 Worked example: explain the prediction

**Reference:** $v_f = v_i + g\,\Delta t$.
Downward positive; g = **10 m/s²**; gravity only.

**Worked reasoning:** A ball dropped from rest has downward speeds **0**, **10**, **20 m/s** at **0**, **1**, **2 s** in ideal free fall with g=**10 m/s²**. Equal times give equal velocity changes.

**Your turn:** 1. **Sketch its velocity-time graph** with downward positive.
2. **Explain why the positions** are not equally spaced.

**Sentence starter:** I think … . I can point to …, which shows …, so … .




**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Two balls of different mass are released together in a vacuum from the same height. **Compare acceleration** and arrival time.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A ball is released from rest in a drop simulation. Ignore air resistance and use g = **10 m/s²**. **Complete the table**, then **describe how the velocity changes** each second.

**Sentence starter:** I notice … . This supports … because … .

| Time (s) | Downward velocity (m/s) |
|---|---|
| 0 | 0 |
| 1 | ____ |
| 2 | ____ |
| 3 | ____ |

**Marzano check - circle:** ① ② ③

# Free fall

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the statement.

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A fixed-mass cart has acceleration **4 m/s²** under **8 N** net force. **Find acceleration** when net force becomes **12 N**. Explain.
**Use the table** to keep the controlled and changed quantities distinct.

**Word help:** net force = the vector sum of all forces on one object

**Sentence starter:** I think … . I can point to …, which shows …, so … .

| Trial | Mass | Net force | Acceleration |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** ① ② ③

## 07 Exit check

A dropped ball falls from rest for **4 s** in an air-free simulation.
1. Use g = **10 m/s²**.
2. **Find its final velocity** and **add a correctly directed arrow** to the diagram.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Dropped ball**

State your positive direction.

Velocity:

Acceleration / force:

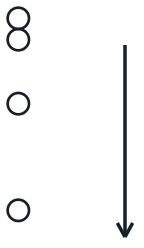
**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

**Draw three equal-time snapshots of** a dropped ball. Then **draw a heavier ball released** with the same setup in a vacuum. **Compare the patterns**.

**Sentence starter:** Both … . They differ in … because … .

Equal time intervals



**Marzano check - circle:** ① ② ③